SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Deep Learning COURSE CODE: MLA04

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COURSE OUTCOME COVERED

*CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)*

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QUESTIONS: 1

Duration: 1 Hour
Total Marks:50

Annexure A

Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

**Solution:**

**Introduction**

A Neural Network is a machine learning model that learns patterns from training data and uses them to make predictions on new, unseen data. The performance of a regression model is commonly measured using **Mean Squared Error (MSE)**. MSE calculates the average of the squared differences between the actual values and predicted values. A lower MSE indicates better prediction accuracy.
In this problem:

- **Training MSE = 0.85**
- **Testing MSE = 5.20**

The training error is very low, while the testing error is much higher. This indicates that the model performs well on the training data but fails to generalize to new data. This is a common issue in machine learning known as **overfitting**

**Step 1: Error Identification**

The large difference between training and testing MSE indicates that the neural network has learned the training data too well but cannot make accurate predictions for unseen data.
**Observation**

| Dataset | MSE |
|---|---|
| Training Data | 0.85 |
| Testing Data | 5.20 |

Since the testing error is much larger than the training error, the model is not generalizing properly.

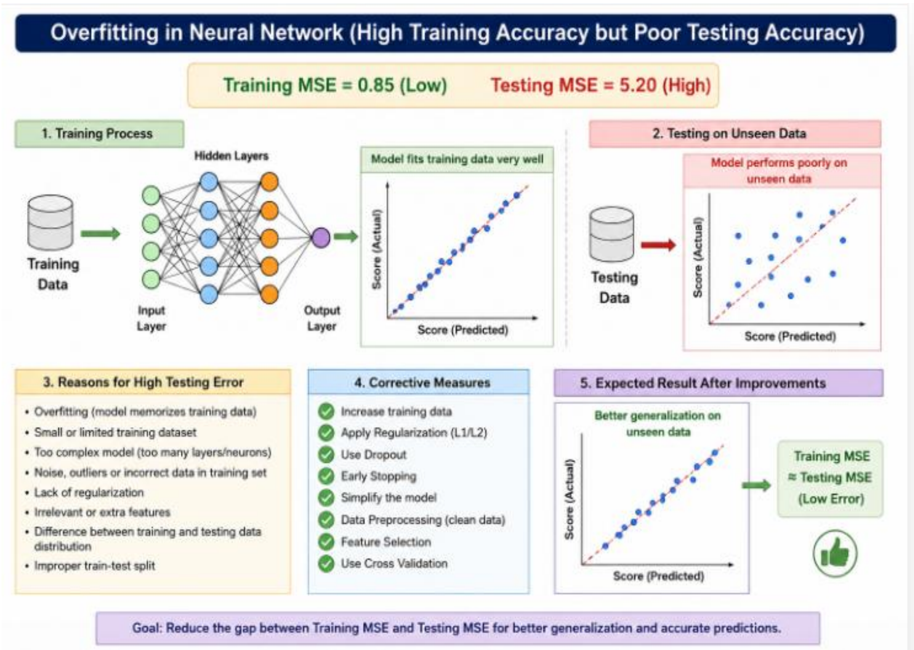
**Main Error Identified**

**Overfitting**

Overfitting occurs when the neural network memorizes the training examples instead of learning the underlying relationship between input and output.

As a result:

- Training predictions become highly accurate.
- Testing predictions become inaccurate.
- The model performs poorly in real-world situations.




**Step 2: Analysis of Possible Reasons**

Several factors may have caused this large difference in error.

**1. Overfitting**

This is the most likely reason.

The neural network has learned unnecessary details and noise present in the training dataset.

Instead of learning general patterns, it memorizes the training examples.

**Impact**

- Very low training MSE
- High testing MSE
- Poor prediction on new student data

**2. Small Training Dataset**

If the dataset contains only a limited number of student records, the model cannot learn enough variations.

The network becomes highly dependent on the available data.

**Impact**

- Limited learning
- Poor generalization
- High testing error

**3. Complex Neural Network**
A neural network with many hidden layers or too many neurons has a large number of parameters.
When the model is more complex than necessary, it starts memorizing the training data.
**Impact**
- Increased overfitting
- Reduced testing performance

**4. Noise in Training Data**
The training dataset may contain:
- Incorrect student scores
- Missing values
- Data entry mistakes
- Outliers

The model attempts to learn these incorrect values.
**Impact**
- Wrong learning
- Reduced prediction accuracy
- Higher testing MSE

**5. Lack of Regularization**
Regularization techniques prevent the neural network from becoming overly complex.
If regularization is not used, the model learns every detail of the training data.
**Impact**
- Larger model weights
- Memorization
- Poor testing performance

**6. Poor Feature Selection**
Some input features may not contribute to predicting student scores.
For example:
- Student ID
- Registration Number
- Random values

These unnecessary features confuse the neural network.
**Impact**
- Increased complexity
- Lower accuracy
- Higher testing error

**7. Distribution Difference**
The testing dataset may differ from the training dataset.
Example:
Training data contains scores of students from one department.
Testing data contains scores from another department with different performance levels.
**Impact**
- Model cannot adapt
- High prediction error

**8. Improper Data Splitting**
If training and testing data are not divided randomly, one dataset may become easier than the other.
Example:
Training data contains average-performing students.

Testing data contains mostly weak or top-performing students.
**Impact**
- Biased learning
- High testing MSE

**Step 3: Error Analysis and Reasoning**
The neural network performs extremely well during training because it has already seen the training data many times.
However, during testing, the model receives completely new student records.
Since the network has memorized the training examples rather than learning general relationships, it cannot predict these new records accurately.
The difference can be calculated as:
**Difference in MSE**
= Testing MSE − Training MSE
= 5.20 − 0.85
= **4.35**
This large difference clearly indicates poor generalization ability.
A well-trained neural network should have training and testing errors that are relatively close to each other.

**Step 4: Correction Strategies**
Several corrective measures can improve the model.
**1. Increase Training Data**
Collect more student records from different classes, semesters, and departments.
Benefits:
- Better learning
- Improved generalization
- Lower testing error

**2. Apply Regularization**
Regularization prevents the model from learning unnecessary details.
Common methods:
- L1 Regularization
- L2 Regularization
- Weight Decay

Benefits:
- Reduces overfitting
- Improves testing accuracy

**3. Use Dropout**
Dropout randomly removes some neurons during training.
This prevents the network from depending too much on particular neurons.
Benefits:
- Better learning
- Less overfitting
- Improved generalization

**4. Early Stopping**
Monitor the validation error during training.
Stop training when validation error starts increasing.
Benefits:
- Prevents over-training

- Reduces overfitting

**5. Simplify the Neural Network**
Reduce:
- Number of hidden layers
- Number of neurons
- Total parameters

Benefits:
- Simpler model
- Better testing performance

**6. Data Preprocessing**
Improve data quality by:
- Removing missing values
- Eliminating duplicate records
- Correcting incorrect entries
- Handling outliers
- Normalizing data

Benefits:
- Better learning
- Higher prediction accuracy

**7. Feature Selection**
Keep only important features such as:
- Attendance
- Previous semester marks
- Assignment scores
- Internal assessment marks
- Study hours

Remove irrelevant features.
Benefits:
- Less complexity
- Better accuracy

**8. Cross Validation**
Use techniques like **K-Fold Cross Validation**.
Benefits:
- Better evaluation
- Stable model performance
- Reliable testing accuracy

**Step 5: Interpretation of Results**
The obtained MSE values indicate that the model has learned the training data very well but cannot accurately predict unseen student scores.
The high testing MSE shows that the neural network lacks good generalization capability.
If no corrections are applied:
- Prediction accuracy will remain poor.
- Student performance predictions will become unreliable.
- The model may not work effectively in real-world educational applications.

After applying corrective measures such as regularization, dropout, early stopping, better preprocessing, feature selection, and collecting more data, the gap between training and testing MSE will reduce. This results in a more accurate and reliable neural network.

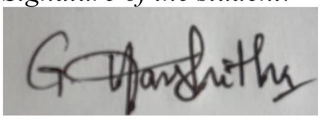
**Conclusion**

The neural network shows a **training MSE of 0.85** and a **testing MSE of 5.20**, indicating a significant performance gap caused mainly by **overfitting**. Other contributing factors include insufficient training data, excessive model complexity, noisy data, poor feature selection, lack of regularization, distribution differences, and improper data splitting. By applying techniques such as regularization, dropout, early stopping, simplifying the network, improving data quality, selecting relevant features, and using cross-validation, the model can generalize better to unseen data. As a result, the testing MSE will decrease, prediction accuracy will improve, and the neural network will become more reliable for predicting student scores.

*Code of Conduct:*

*I (G.Harshitha / Reg No:192372384) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

*Signature of the student:*



RUBRICS FOR CALCULATION

| **Criteria** | **Weightage** | **Level 4 – Exemplary (5)** | **Level 3 – Proficient (4)** | **Level 2 – Developing (2-3)** | **Level 1 – Beginning (0-1)** |
|---|---|---|---|---|---|
| Error Identification | 30% | Correctly identifies all errors and their causes | Identifies most errors with minor omissions | Identifies some errors but misses key issues | Unable to identify errors |
| Error Analysis & Reasoning | 30% | Provides clear and logical explanation of why errors occurred | Reasoning mostly corrects with minor gaps | Limited reasoning and partial explanation | No meaningful analysis |
| Correction Strategy | 25% | Suggests appropriate corrections with justification | Suggests correct corrections with minor gaps | Partially correct corrections | Incorrect or no corrections |
| Interpretation of Results | 15% | Clearly explains impact of errors on model performance/results | Basic interpretation with minor omissions | Limited interpretation | No interpretation provided |